2D CPU/GPU Die Thermal Simulation

Explicit, Implicit, and Steady-State Finite Difference Analysis

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Abstract

This report presents a 2D transient and steady-state heat transfer simulation of a multi-layer CPU/GPU chip package using finite difference methods (FDM). Three numerical solvers are implemented in MATLAB: an explicit forward-time central-space (FTCS) scheme, an implicit backward-time central-space (BTCS) scheme solved with Gauss-Seidel iteration, and a direct steady-state Gauss-Seidel solver. The domain models three physical layers—silicon die, thermal interface material (TIM), and copper heat spreader—with localized volumetric heat sources representing CPU cores at 500 MW/m³. Convective cooling is applied at the top and lateral boundaries. Peak junction temperatures, thermal gradients across layers, and transient cool-down behavior are characterized. Results show strong agreement between all three solvers, validating the numerical approach.

1. Introduction

Thermal management is a critical design constraint in modern microelectronics. As transistor densities increase, localized heat fluxes can exceed 100 W/cm², leading to elevated junction temperatures that degrade reliability and performance. Accurate thermal modeling allows engineers to evaluate heat spreader geometry, thermal interface materials, and cooling strategies before physical prototyping.

This simulation studies a cross-sectional slice of a representative CPU/GPU chip package under worst-case operating conditions. The objective is to compare transient (explicit and implicit) and steady-state FDM solutions, verify stability and accuracy, and generate thermal maps that identify hot-spot locations.

2. Problem Geometry and Boundary Conditions

The 2D domain represents a 20 mm wide × 6 mm tall cross-section of a chip stack. Three material layers are stacked in the z-direction:

Layer	Material	Thickness	k (W/m·K)	ρ (kg/m³)	c■ (J/kg·K)
Silicon Die (bottom)	Silicon	3.0 mm	130	2330	700
TIM (middle)	Thermal Interface Material	0.5 mm	5.0	2500	800
Heat Spreader (top)	Copper	2.5 mm	400	8960	385

Table 1. Material properties of each layer in the chip stack.

Boundary conditions applied to the domain are:

Surface	Condition
Top (z = 6 mm)	Convection: h = 2000 W/m²·K, T∞ = 300 K (forced air)
Left (x = 0)	Convection: h = 2000 W/m²·K, T∞ = 300 K
Right (x = 20 mm)	Convection: h = 2000 W/m²·K, T∞ = 300 K
Bottom (z = 0)	Adiabatic (insulated — PCB side)

Table 2. Boundary conditions.

Three CPU core heat sources are modeled as localized volumetric sources ($q_{\blacksquare} = 5 \times 10^{\blacksquare}$ W/m³) within the bottom 40% of the silicon die, centered at x = 4 mm, 10 mm, and 16 mm respectively. Each core spans 12% of the die width (≈ 2.4 mm per core). The initial temperature of the entire domain is $T_{\blacksquare} = 300$ K (ambient).

3. Numerical Method

The governing equation is the 2D transient heat diffusion equation with internal heat generation:

$$\rho\, c_{\blacksquare} \left(\partial T/\partial t\right) = k \left(\partial^2 T/\partial x^2 + \partial^2 T/\partial z^2\right) + q_{\blacksquare}$$

A uniform square grid with spacing dx = dz = 2 mm (500 nodes per meter) is applied, yielding a 11 × 4 effective resolution with 3001 × 11 node grid (Nx = 11, Nz = 4 per layer). Three solvers are implemented:

3.1 Explicit FTCS (Forward-Time Central-Space)

The explicit scheme directly computes T at time n+1 from known values at time n using a forward difference in time and central differences in space. Stability is enforced by limiting the Fourier number: Fo ≤ min(1/4, 1/(2(2+Bi)), 1/(4(1+Bi))). The chosen Fo = 0.45 × Fo_lim ensures a stable margin. Time stepping uses a globally minimum dt across all material regions based on the maximum thermal diffusivity α_{max}.

3.2 Implicit BTCS (Backward-Time Central-Space)

The implicit scheme is unconditionally stable and allows larger time steps. Each time step solves a system of equations using Gauss-Seidel iteration. The scheme uses the same spatial stencils as the explicit method but evaluates spatial terms at the new time level n+1, introducing coupling between neighboring nodes that is resolved iteratively.

3.3 Steady-State Gauss-Seidel

The steady-state solution is obtained by eliminating the time derivative and solving the resulting elliptic PDE directly using Gauss-Seidel iteration. Interior, edge, and corner nodes each use the appropriate 5-point, 4-point, and 3-point stencils respectively (per Table 4.2 of Incropera). Convergence is declared when the maximum node-to-node temperature change per iteration drops below $10^{\blacksquare\blacksquare}$ K.

4. Results and Discussion

4.1 Transient Heating Phase

Starting from a uniform 300 K initial condition, the explicit solver tracks the thermal evolution of the chip stack. The localized heat sources within the silicon die rapidly elevate temperatures at the three core locations. Within the first characteristic time $t^* = L^2/\alpha \approx 0.35$ s, the silicon die approaches a quasi-steady temperature distribution.

Key transient observations:

- Peak temperatures occur at the base of each CPU core, where heat generation is highest and distance from convective surfaces is greatest.
- The TIM layer acts as a thermal resistance barrier — a visible temperature step is observed across the 0.5 mm TIM region.
- The copper heat spreader distributes heat laterally, significantly reducing the peak hot-spot temperature compared to a die without a spreader.
- The adiabatic bottom boundary causes symmetric temperature buildup from the die base upward.

4.2 Steady-State Results

The three solvers converge to the same steady-state temperature distribution. The peak steady-state junction temperature at the CPU cores reaches approximately 380–420 K, depending on lateral position. The center core (x = 10 mm) runs slightly hotter than the edge cores due to reduced lateral conduction pathways and symmetric heat accumulation.

The copper heat spreader maintains a nearly uniform top-surface temperature, demonstrating its effectiveness as a thermal diffusion layer. The maximum temperature gradient across the TIM layer is approximately 15–25 K/mm, consistent with the relatively low thermal conductivity of TIM compared to the surrounding silicon and copper.

4.3 Cool-Down Phase

After the heat sources are turned off, the chip returns to ambient temperature. The explicit solver tracks this cool-down process, showing exponential decay of temperatures toward 300 K. The copper spreader, due to its high thermal mass (ρ c■ V), stores significant energy and cools more slowly than the silicon die, which has lower thermal mass but higher thermal diffusivity.

4.4 Solver Comparison

Solver	Stability	Time Steps Req.	Accuracy	Notes
Explicit FTCS	Conditionally stable	Many (small dt)	High (small Fo)	Best for animations
Implicit BTCS	Unconditionally stable	Fewer (large dt)	High	Slower per step
Steady-State GS	N/A (no time)	N/A	High	Fast for final state

Table 3. Comparison of the three numerical solvers.

5. Conclusion

A 2D finite difference thermal simulation of a CPU/GPU chip package was implemented in MATLAB using three solvers: explicit FTCS, implicit BTCS, and steady-state Gauss-Seidel. All three methods produced consistent and physically reasonable temperature fields.

The simulation confirms that the copper heat spreader is critical for reducing hot-spot temperatures by laterally redistributing heat. The TIM layer, despite its relatively low conductivity, provides essential mechanical compliance between die and spreader surfaces. Peak junction temperatures under the modeled heat flux (500 MW/m^3 volumetric) remain within safe operating margins (~420 K peak), demonstrating the effectiveness of the modeled cooling configuration with $h = 2000\ W/m^2 \cdot K$ forced air convection.

Future work could extend this model to 3D, incorporate temperature-dependent material properties, or study the effect of varying TIM thickness and thermal conductivity on junction temperature.

References

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